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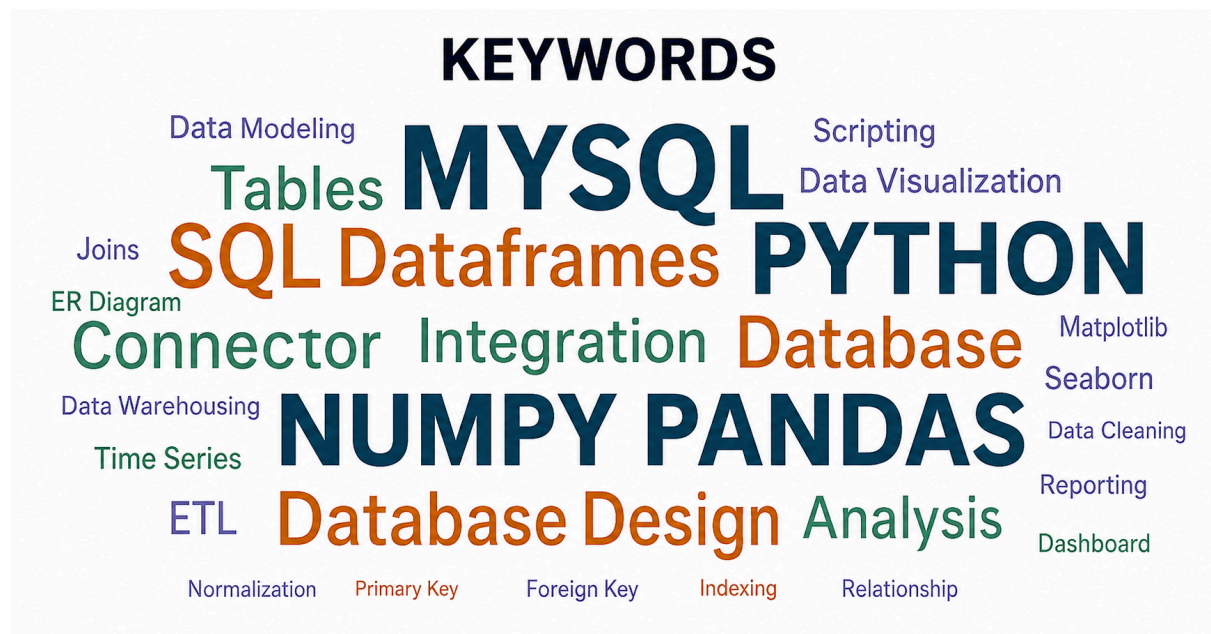


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Abstract

This report is about changing a database that uses paper into a digital database system that uses MySQL. The digital database system is called a Relational Database Management System. We use this system to store and manage data.

We also show how to look at the data and make pictures with it using Python. The Python tools we use are NumPy, Pandas and Matplotlib. We use these tools to analyze and visualize the data, from the database system that uses MySQL.



Acknowledgement

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Introduction

In an increasingly fast-paced construction and engineering industry, relying on fragmented spreadsheets to manage projects, equipment, and personnel records has become a major source of inefficiency and error ([Data Management in the Era of Digitalization, 2019](#)). This project addresses that problem by transforming the disorganized records of BuildCorp International, a Kathmandu-based engineering firm, into a properly structured and centralized relational database system capable of supporting project tracking, equipment hire management, and future analytics.



Figure 1: Steps to Transform Spreadsheet-Based Records into a Centralized Database

To achieve this transformation, the project is divided into four key stages:

1. Normalization
2. Entity-Relationship Diagram
3. SQL Implementation
4. Data Visualization

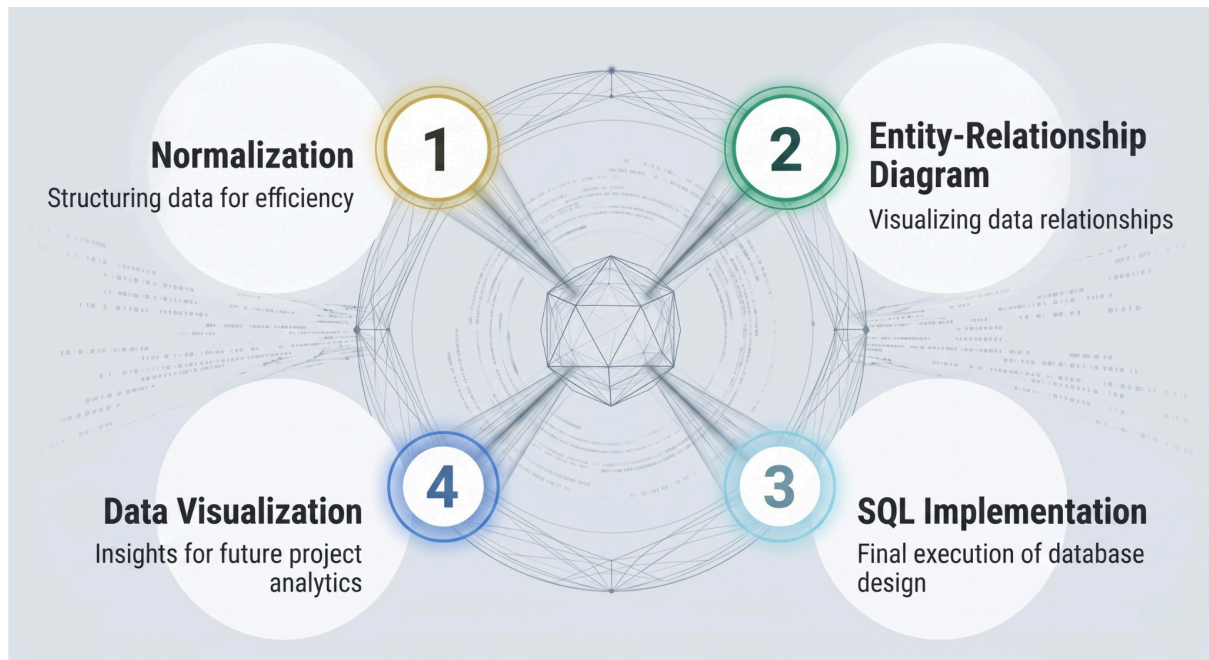


Figure 2: Four Strategies for BuildCorp's Database Transformation

Task 1: Design

Normalization Process

Database normalization is the process of structuring a relational database in accordance with a series of normal forms to reduce data redundancy and improve data integrity. It was first proposed by British computer scientist Edgar F. Codd as part of his relational model. ([Database Normalization, n.d.](#)). Rather than storing all information in one large, repetitive table, the data is broken down into smaller, logically connected tables, each representing a single concept or entity. For this project, the BuildCorp tables are normalized progressively until they satisfy the requirements of Third Normal Form (3NF).

The normalization process generally moves through four recognizable stages:

- 1. Unnormalized Form (UNF)
- 2. First Normal Form (1NF)
- 3. Second Normal Form (2NF)
- 4. Third Normal Form (3NF)

Unnormalized Form (UNF)

Table 1: Project Master Contact Registry

Proj_ID	Project_Name_&_Owner	Contact_Person_Details	Contact_Type	Proj_Dates
P-901	Bagmati Bridge (Govt. Nepal)	Rajesh Hamal (r.hamal@email.com)	Lead Engineer	2026-01-01 to 2027-12-31
P-902	Pokhara Airport (Civil Aviation)	Sita Rai (sita.r@email.com)	Site Manager	2026-02-15 to 2028-06-30
P-903	Kathmandu Metro (KMC)	Binod Das (b.das@email.com)	Lead Engineer	2026-03-01 to 2029-01-01

Table 2: Equipment Catalog Multi-Categories

Equip_ID	Machine_Name	Category_List	Current_Status	Status_Notes
E-101	Caterpillar Excavator 320	Heavy, Earthmoving, Hydraulic	Active	Fully Operational
E-102	Tower Crane XJ	Heavy, Lifting	Maintenance	Returning March 2026
E-103	Volvo Haul Truck	Heavy, Transport	Active	New Tires Installed

Table 3: Complex Hire Rate Tiers

Rate_ID	Equip_Linked	Daily_Rate	Weekly_Rate	Hire_Period_Notes
HR-55	E-101	150	900	Includes Operator Fuel
HR-60	E-102	500	3000	Dry Hire Only
HR-65	E-103	250	1400	Weekly discount applied

Table 4: Global Contact Type Mapping

Type_ID	Type_Name	Security_Level	Dept_Head
T-1	Lead Engineer	Level 4	Dr. Binod
T-2	Site Manager	Level 3	Er. Suman
T-3	Client Rep	Level 2	Ms. Thapa

Table 5: Project HR Testimonials

HR_Rec_ID	Project_ID	HR_Name_&_Role	Testimonial_Text
REC-01	P-901	Rajesh Hamal (Lead Engineer)	"Excellent hydraulic power on site."
REC-02	P-902	Sita Rai (Site Manager)	"Crane maintenance was slow."
REC-03	P-901	Binod Das (Consultant)	"Project ahead of schedule."

Table 6: Equipment Category Bridge (Raw Log)

Bridge_ID	Equipment_ID	Category_ID_Mapping	Last_Audit_Date
B-1	E-101	CAT-01, CAT-04, CAT-07	2025-12-01
B-2	E-102	CAT-01, CAT-05	2025-11-15
B-3	E-103	CAT-01, CAT-09	2026-01-10

Table 7: Master Booking Calendar Table

Cal_ID	Booking_Date	Project_ID	Equip_ID	Invoice_Status
C-1001	2026-02-15	P-901	E-101	Confirmed: INV-442
C-1002	2026-02-16	P-901	E-101	Pending: NO_INV
C-1003	2026-02-17	P-902	E-103	Confirmed: INV-445

Table 8: Project Equipment Financials

Proj_Equip_ID	Project_ID	Equip_ID	Booking_Rate_Applied	Duration_Days
PE-55	P-901	E-101	145	10
PE-56	P-902	E-103	250	5
PE-57	P-901	E-103	230	15

First Normal Form (1NF)

Data Management in the Era of Digitalization. (2019, November 7). Research Gate.
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